

IPSec IKEv2 Route-Based

VPN Site-to-Site basada en Enrutamiento con IKEv2

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1. Objetivo

Configurar una VPN Site-to-Site punto a punto basada en enrutamiento utilizando IPsec con IKEv2. Se crea una interfaz virtual Tunnel0 protegida con un perfil IKEv2, y se enruta el tráfico hacia la LAN remota a través de esa interfaz sin necesidad de ACLs.

Topología de Red

La topología consiste en dos routers peers (R1 y R2) conectados a través de un router ISP que simula internet, con hosts Linux en cada LAN para demostrar la conectividad.

 **SCREENSHOT: Insertar screenshot de la topología completa en EVE-NG**

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Rol
R1	Fa0/0	10.7.80.1/30	WAN → ISP
R1	Fa1/0	192.168.7.1/24	LAN
ISP	Fa0/1	10.7.80.2/30	WAN → R1
ISP	Fa0/0	10.7.81.1/30	WAN → R2
R2	Fa0/0	10.7.81.2/30	WAN → ISP
R2	Fa1/0	192.168.80.1/24	LAN
Linux1	ens3	192.168.7.2/24	Host LAN R1
Linux2	ens3	192.168.80.2/24	Host LAN R2

Dispositivo	Interfaz Tunnel	IP Tunnel
R1	Tunnel0	172.16.78.1/30
R2	Tunnel0	172.16.78.2/30

2. Parámetros VPN

Parámetro	Valor
Tipo de VPN	IPSec IKEv2 Route-Based
IKEv2 Proposal — Cifrado	AES-CBC-128
IKEv2 Proposal — Integridad	SHA-1
IKEv2 Proposal — Grupo DH	Group 2 (1024-bit)
IKEv2 — Autenticación	Pre-Shared Key (PSK)
Fase 2 — Modo	Tunnel
Pre-Shared Key	Yeury0780
Interfaz Tunnel	Tunnel0
Red Tunnel	172.16.78.0/30

3. Configuración

3.1 Configuración R1

```
crypto ipsec profile VPN-PROFILE set transform-set TS set ikev2-profile PROF-  
IKEv2 interface Tunnel0 ip address 172.16.78.1 255.255.255.252 tunnel source  
FastEthernet0/0 tunnel destination 10.7.81.2 tunnel protection ipsec profile  
VPN-PROFILE ip route 192.168.80.0 255.255.255.0 172.16.78.2
```

3.2 Configuración R2

```
crypto ipsec profile VPN-PROFILE set transform-set TS set ikev2-profile PROF-  
IKEv2 interface Tunnel0 ip address 172.16.78.2 255.255.255.252 tunnel source  
FastEthernet0/0 tunnel destination 10.7.80.1 tunnel protection ipsec profile  
VPN-PROFILE ip route 192.168.7.0 255.255.255.0 172.16.78.1
```

4. Verificación y Funcionamiento

4.1 Estado IKEv2 SA

```
Jun 30 01:58:20.051: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on FastEthernet0/0 (not half duplex), with ISP FastEthernet0/1 (half duplex).
R1#
R1#
R1#
R1#
R1#
R1#show crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
-----
2 10.7.80.1/500 10.7.81.2/500 none/none READY

Encr: AES-CBC, keysize: 128, Hash: SHA96, DH Grp:2, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/435 sec

IPv6 Crypto IKEv2 SA
```

El estado READY confirma que IKEv2 está negociado y activo.

4.2 Estado Tunnel0

```
R1# show interface tunnel0
Tunnel0 is up, line protocol is up
Hardware is Tunnel
Internet address is 172.16.78.1/30
MTU 17854 bytes, BW 100 Kbit/sec, DLY 50000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 10.7.80.1 (FastEthernet0/0), destination 10.7.81.2
Tunnel Subblocks:
src-track:
Tunnel0 source tracking subblock associated with FastEthernet0/0
Set of tunnels with source FastEthernet0/0, 1 member (includes iterators), on interface <OK>
Tunnel protocol/transport GRE/IP
```

La interfaz Tunnel0 up/up con 'Tunnel protection via IPSec (profile VPN-PROFILE)' confirma el túnel activo.

4.3 Estado IPsec SA

```
R1#show crypto ipsec sa

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.7.80.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.7.80.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (10.7.81.2/255.255.255.255/47/0)
  current peer 10.7.81.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 17, #pkts encrypt: 17, #pkts digest: 17
    #pkts decaps: 17, #pkts decrypt: 17, #pkts verify: 17
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
```

Los contadores confirman el tráfico cifrado por el túnel IKEv2.

4.4 Demostración de conectividad

```
root@ubuntu22-server:~# ping -c 4 192.168.80.2
PING 192.168.80.2 (192.168.80.2) 56(84) bytes of data.
64 bytes from 192.168.80.2: icmp_seq=1 ttl=62 time=330 ms
64 bytes from 192.168.80.2: icmp_seq=2 ttl=62 time=139 ms
64 bytes from 192.168.80.2: icmp_seq=3 ttl=62 time=169 ms
64 bytes from 192.168.80.2: icmp_seq=4 ttl=62 time=196 ms
```

El ping exitoso demuestra la comunicación cifrada entre las dos LANs con IKEv2 Route-Based.